Reimagining Robot Development with AI and Agentic Workflows

Artificial intelligence is reshaping robotics beyond perception and decision-making by bringing intelligence to the entire development lifecycle. This talk explores how AI can support key stages of robot development - from platform design and sensor integration to perception, planning, and control. It will demonstrate how combining AI with simulation and model-based engineering can shorten development cycles, improve design exploration, and enable increasingly autonomous and capable robotic systems.

The talk also introduces Agentic AI as an emerging approach for orchestrating end-to-end engineering workflows, spanning requirements, architecture and design, modeling, simulation, testing, and deployment. Through examples using MATLAB and Simulink Agentic Toolkits, we will discuss how AI agents can interact with engineering models, simulations, and development tools to assist engineers while supporting human-guided workflows. Together, these advances in AI contribute to the emerging vision of Physical AI: intelligent systems that can perceive, reason, plan, act, and learn within the physical world.

Short Bio:



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Dr. Pranav Lad is a Senior Customer Success Engineer at MathWorks, where he has been a member of the Education Team for over six years, based in Pune, India. He collaborates closely with academic institutions to accelerate research and enhance learning outcomes.

His technical expertise spans robotics, autonomous systems, vibrations, nonlinear dynamics, and artificial intelligence. His current focus is on the modelling and simulation of robotics and autonomous systems, with special emphasis on UAVs and automated driving.

He holds a Ph.D. and M.Tech. in Mechanical Engineering from IIT Bombay, where he also served as a research assistant, engaging with various industries on consulting projects. He earned his bachelor's degree from the University of Pune.